



THE UNIVERSITY *of* EDINBURGH

Assignment 1: Annotated Bibliography

Course Name: LASC10115 Online Experiments for Language Scientists

Word Count: 1,500

Article 1

Pavlick, E., Post, M., Irvine, A., Kachaev, D., & Callison-Burch, C. (2014). The language demographics of Amazon Mechanical Turk. *Transactions of the Association for Computational Linguistics*, 2, 79–92. https://doi.org/10.1162/tac1_a_00167

[+1 Reference included](#)

This study examines MTurk's bilingual population, aiming to enhance the methodology for determining language demographics. Research questions concern the accuracy of self-reported language abilities, the speed of task completion across languages, the value of Turkers' translations for machine translation (MT) systems, and their improvement of MT quality. The experiment involved 1,000 assignments in 100 languages, with 5,281 Turkers earning \$0.15 (word translations) or \$0.70 (sentence translations) per task. Task duration was not reported.

First, participants completed questionnaires on language skills and location. Then, their responses were validated through translation tasks and IP-geolocation. Each worker translated ten L2 words into English, including two control words with established translations. Items were selected from the 10,000 most frequent in Wikipedia's top-viewed articles. Translations matching gold-standard translations from Wikipedia were marked correct, whereas non-matches were re-evaluated. Workers also translated ten sentences from the 100 most-viewed Wikipedia pages in six Indian languages. To deter the use of translation tools, items were converted to images, and participants with over 70% overlap were excluded. Results showed that 13 languages, e.g. Dutch/French/German, yielded quick, accurate task completion. Moreover, crowdsourcing helped develop and enhance MT. The authors recommend selecting in-region participants, using pretests or gold-standard checks, excluding likely machine translations, and tracking response times.

[Summary 4/4](#)

This method has applications in mono- and bilingual settings. For instance, validation checks are also valuable for experiments involving monolinguals, e.g. Hu et al. (2014). Another strength is the release of all data and code, enabling experiment replication. Moreover,

leveraging Wikipedia's multilingual reach and consistent structure appears practical and cost-effective. Nevertheless, on collaborative platforms such as Wikipedia, content and quality might vary, especially for minor languages, which typically have shorter, less detailed entries, resulting in limited vocabulary pools. Additionally, validating language skills via geolocation seems unreliable, as location does not necessarily reflect language proficiency. Focusing on in-region workers risks considerably reducing the number of participants, especially in low-resource languages. Instead, although time-consuming and expensive, oral fluency could be tested by having participants record monologues, e.g. about their families. This also incorporates more effective language assessment than simple translation.



Evaluation 4/5

The recommendations against cheating are particularly relevant to my experiment on UK English speakers' attitudes towards Austrian-accented English. If utilising MTurk, I would restrict participants to British locations and integrate validation tests, e.g. narrative clozes.



Article 2

+1 Reference inc

Lev-Ari, S. (2017). Talking to fewer people leads to having more malleable linguistic representations. *PloS One*, 12(8), 1–12. <https://doi.org/10.1371/journal.pone.0183593>

This online experiment investigates whether individuals with fewer social contacts have more shapeable phonological representations, particularly in categorising /d/ and /t/ with non-standard voice onset times (VOTs), and if they learn specific speakers' speech patterns. 148 US-based English-speaking Turkers participated, receiving \$1.20 for 5-15 minutes.

After completing a social network questionnaire, participants engaged in picture selection trials with either manipulated /d/ or /t/ conditions. Audio stimuli included 20 phrases with /d/ or /t/ and 40 fillers. For critical phrases, ambiguous tokens with 24ms-VOT replaced /d/ or /t/, depending on the condition. Participants then completed a phoneme categorisation task, choosing between 'dean'/'teen' in two conditions: same/new speaker. Stimuli, recorded by the initial and a new speaker, included ambiguous or clear /d/ or /t/ sounds with various VOTs (5ms/10ms/15ms/20ms/25ms/30ms/35ms/50ms/80ms). Responses to critical items (15-35ms) were analysed regarding shifts of perceptual boundaries. Results indicated that participants with smaller social networks have more malleable linguistic representations and adapted their representation of phonological categories more readily to atypical input, with perceptual learning largely influenced by manipulated /d/. The effect of social network size is not related to differing abilities in learning speakers' speech patterns but to the likelihood of generalising learned patterns. Lev-Ari suggests that individuals with fewer interlocutors may be more significant for linguistic innovations.

Summary 4/4

The paper's strengths include its transparency about limitations and future research directions. The method's design is simple, facilitating participation but encouraging rushed responses. This issue is compounded by the payment structure; participants taking longer

effectively earn less (\$1.20/15 minutes = \$4.80/hour), potentially reducing attention to detail.



Improved compensation and additional attention checks could enhance data quality. Moreover, self-reported questionnaires entail unreliable responses. Using more effective methods, e.g. diaries (cf. Fu, 2007), though costly and time-consuming, could boost accuracy. Imprecise estimates would only be acceptable with a large sample, but the study has a small one, limiting generalisability.



Counteracting unreliability by trimming the estimates seems inappropriate since this may increase false positives. Further drawbacks are the speculative interpretation of findings, e.g. the role in linguistic change, without solid evidence, and the disregard for uncontrolled influences on malleability, e.g. age.



Evaluation 4/5

I found reviewing scientific implementations of audio stimuli helpful for designing my experiment. Like Lev-Ari, I will employ voice recordings, instruct participants to use high-quality headphones, and include a demographic questionnaire.

Article 3

+1 Reference included

Dragojevic, M., & Goatley-Soan, S. (2022). Americans' attitudes toward foreign accents: Evaluative hierarchies and underlying processes. *Journal of Multilingual and Multicultural Development*, 43(2), 167–181. <https://doi.org/10.1080/01434632.2020.1735402>

Using a verbal-guise technique, this study inspects Americans' attitudes towards nine non-Anglo accents (Arabic/Farsi/French/German/Hindi/Hispanic/Mandarin/Russian/Vietnamese) and standard American English (SAE), focusing on status and solidarity judgements linked to social categorisation and processing fluency. Another aim was to identify a hierarchy among these varieties. Conducted online via MTurk, the experiment enlisted 254 workers, receiving \$5 for 45 minutes. Participants were instructed to evaluate 'personalities', presumably to avoid prejudice.

Following a sound check, each participant listened to 20 of 40 recordings, featuring one female and one male speaker per accent reading a neutral, fictional text, which potentially controls gender and content bias. Recordings played automatically without a skip option, which seems useful to stop participants from button-mashing, ensuring engagement and response accuracy. Then, participants rated speakers on processing fluency, four status traits, e.g. 'competent', and four solidarity traits, e.g. 'warm', with 7-point scales. Even-numbered scales could have promoted more definitive answers by eliminating the neutral option. Social categorisation was assessed through an open-ended question about speakers' nationalities. Results showed that SAE speakers received the best ratings, while certain accents, e.g. German, scored higher than others, e.g. Arabic. This hierarchy correlated with processing fluency and social categorisation.

Summary 4/4

In general, subjective descriptors are questionable, as participants may interpret terms like 'competent' differently, implying inconsistent responses. Researchers could clarify the descriptors by providing concrete definitions or pilot-testing them. This is worth considering

 6

for my project. Nevertheless, the method appears comprehensive, versatile, and adaptable to other topics, e.g. gendered language perceptions. Utilising a 'cover' to prevent bias also seems effective (cf. Loy & Smith, 2021). However, compensation was relatively low, raising ethical concerns, and utilising a single, context-free passage may limit generalisability. Access to the text was not provided, hindering readers from replicating or comprehending the experiment. Moreover, individual differences influencing attitudes, e.g. education, were overlooked.

This study is highly relevant to my second assignment. I plan to use a similar verbal-guise technique with UK participants, focusing on Austrian-accented English, and adopt the 'cover' strategy, using fake scenarios, e.g. assessing job suitability, to mask the focus on attitudes (cf. Dalton-Puffer et al., 1997). I would also add a sound check and use an unbiased text, like Dragojevic and Goatley-Soan, although I would opt for a more authentic, factual one, e.g. a weather forecast.



Evaluation 5/5

Article 4

Dalton-Puffer, C., Kaltenboeck, G., & Smit, U. (1997). Learner attitudes and L2 pronunciation in Austria. *World Englishes*, 16(1), 115–128. <https://doi.org/10.1111/1467-971X.00052>

+1 Reference included

This on-site study examines Austrian learners' attitudes towards native and non-native English accents and their impact on the attainment as well as the desirability of native-like pronunciation. Using a modified matched-guise technique, the authors investigated preferences for Received Pronunciation (RP), near-RP, General American, Austrian-accented British English, and Austrian-accented American English among 132 university English students, participating during their courses. The stimuli included five female speakers, one per accent, reading a neutral passage, framed as an audition for an audiobook, making the test situation more plausible.

After each stimulus, students evaluated the speakers regarding 12 attributes, e.g. intelligence, using a 5-point scale, with 'not applicable' as the zero option to reduce middle-value tendency. Participants also assessed whether the speaker would be a suitable radio presenter or friend, then identified speakers' nationalities and provided information about themselves, e.g. stays abroad. Overall, native speakers were rated higher than non-natives – RP highest and Austrian-accented British English lowest. Respondents preferred accents they were most familiar with, particularly RP, common in Austrian schools. The authors note that most learners struggle to attain this preferred pronunciation, suggesting that phonology learning is not solely driven by attitudes. This claim would benefit from supporting data about the learners' lacking achievement. Stating that attitudes alone do not dictate phonology learning also seems slightly redundant. Investigating the extent of their influence would be more insightful, though this requires a larger study incorporating socio-psychological or physical factors.

9

Summary 4/4

Several weaknesses emerged. First, the study does not clarify whether participation was voluntary, raising ethical questions. Second, the setting is unspecified, leaving concerns about sound quality or peer influence. Third, using only female voices may introduce gender bias. Utilising both genders (cf. Dragojevic & Goatley-Soan, 2022) would enable generalisation across speaker gender. Fourth, focusing on university students limits generalisability. Future studies could include varied groups, e.g. vocational school students, for broader insights.

Strengths include using multiple independent variables, e.g. stays abroad, and a straightforward method, adaptable for online data collection. For my project, I strive to reproduce its structure in jsPsych, e.g. with audio-slider-response plugins. The attached questionnaire, descriptors, and text facilitate replication. However, the experiment might not be suitable for crowdsourcing websites. Testing educational platforms may ensure more appropriate participants.

Evaluation 4/5

Works Cited

- Fu, Y. (2007). Contact diaries: Building archives of actual and comprehensive personal networks. *Field Methods*, 19(2), 194–217. <https://doi.org/10.1177/1525822X06298590>
- Hu, C., Resnik, P., & Bederson, B. B. (2014). Crowdsourced monolingual translation. *ACM Transactions on Computer-Human Interaction*, 21(4), 1–35. <https://doi.org/10.1145/2627751>
- Loy, J. E., & Smith, K. (2021). Speakers align with their partner's overspecification during interaction. *Cognitive Science*, 45(12), e13065. <https://doi.org/10.1111/cogs.13065>

List of Abbreviations

L2	Second language
ms	Millisecond
MT	Machine translation
MTurk	Amazon Mechanical Turk
RP	Received Pronunciation
SAE	Standard American English
UK	United Kingdom
VOT	Voice onset time

GRADEMARK REPORT

FINAL GRADE

93/100

GENERAL COMMENTS

Entry 1: 9/10

Entry 2: 9/10

Entry 3: 10/10

Entry 4: 9/10

Total: 37/40, 93%

A really exceptional piece of work, congratulations - I am not sure I could do such a good job myself. Once these are de-anonymised I will be contacting you to ask if I can use this as an example bibliography for future years.

PAGE 1

PAGE 2

QM

+1 Reference included

Reference provided in APA (or similar)

QM

Summary 4/4

Very clear summary, particular useful and/or easy to follow and/or well-expressed.

Additional Comment

It's actually so much more detailed than the other summaries I have read so far that I assumed you must be way over the word count and checked! Well done on expressing so much in such a short space.

PAGE 3

QM

Evaluation 4/5

Very good evidence of thoughtful critical engagement, e.g. very clear and detailed thoughts on a single issue or several solid points raised, perhaps covering the paper on its own merits and in connection to the rest of the literature, and/or unusually clearly expressed.



Comment 1

A bunch of good points here but this is probably the best.



Comment 2

Good plan

PAGE 4

QM

+1 Reference included

Reference provided in APA (or similar)

QM

Summary 4/4

Very clear summary, particular useful and/or easy to follow and/or well-expressed.

PAGE 5



Comment 3

Is this a problem though - could one argue this is more like real time-pressured speech perception?



Comment 4

Nice



Comment 5

Good stuff

QM

Evaluation 4/5

Very good evidence of thoughtful critical engagement, e.g. very clear and detailed thoughts on a single issue or several solid points raised, perhaps covering the paper on its own merits and in connection to the rest of the literature, and/or unusually clearly expressed.

PAGE 6

QM

+1 Reference included

Reference provided in APA (or similar)

QM

Summary 4/4

Very clear summary, particular useful and/or easy to follow and/or well-expressed.



Comment 6

Good point

PAGE 7



Comment 7

Any solutions you can think of for this, or is it inherent to this kind of experiment?

QM

Evaluation 5/5

Exceptional critical evaluation, suggesting very careful thought about the paper. E.g. unusually detailed or insightful comments or connections drawn, and unusually well expressed.

Additional Comment

Really great work



Comment 8

All excellent ideas

PAGE 8

QM

+1 Reference included

Reference provided in APA (or similar)

QM

Summary 4/4

Very clear summary, particular useful and/or easy to follow and/or well-expressed.



Comment 9

Slightly harsh but a good point!

PAGE 9



Comment 10

I would assume it!



Comment 11

This is probably the weakest of your evaluative points, simply because it applies so widely - can you make an argument why this is a particular problem for this kind of research question?



Comment 12

Also a very sensible idea.

QM

Evaluation 4/5

Very good evidence of thoughtful critical engagement, e.g. very clear and detailed thoughts on a single issue or several solid points raised, perhaps covering the paper on its own merits and in connection to the rest of the literature, and/or unusually clearly expressed.

PAGE 10
